IGIS
INSTITUTE OF GEOGRAPHICAL INFORMATION SYSTEMS

Geological Survey of Pakistan



INTERNATIONAL TRAINING WORKSHOP

# Remote Sensing *for Disaster Management*

From satellite constellations to UAV systems: an integrated observation chain for emergency response and loss assessment

**18–20 November 2026**
IGIS, SCEE, NUST · Islamabad

*Programme booklet*

# Two scales. One observation chain.

During a monsoon flood, responders need two answers at once: *how far has the water spread, and who is affected?* Then: *which embankments breached, which bridges failed, how many homes are lost?*

> No single sensing platform can address both. Used in a deliberate sequence, satellites and UAVs create an observation chain far more effective than either alone.

Over two and a half days you will work through that chain, using openly licensed satellite data, browser-based cloud computing and standard commercial UAVs, all reproducible with equipment ordinarily available to a Pakistani institution.

### BY THE CLOSE OF DAY 3 YOU WILL HAVE PRODUCED

1. **A flood extent map** from Sentinel-1 SAR
2. **An exposure table** of population, cropland and roads
3. **A UAV orthomosaic and DSM** from imagery you helped fly
4. **A damage assessment** quantified from that model
5. **A rapid situation brief** for an emergency operations centre

PROGRAMME AT A GLANCE

# From the wide view to the close-up.

01

WEDNESDAY 18 NOVEMBER · 08:30–17:00

## Foundations, wide-area assessment & mission design

Modules 1–3 · Lab 0 · Lab 1A/1B flood mapping and exposure · Lab 2 flight planning

02

THURSDAY 19 NOVEMBER · 08:30–17:00

## Field operations & UAV data processing

Field flights at the NUST field site · Lab 3A photogrammetry · Lab 3B 3D modelling and damage quantification

03

FRIDAY 20 NOVEMBER · 09:00–12:15

## Integration, case studies & rapid-response simulation

Module 4 · three Pakistani case studies · capstone simulation · closing and certificates

| 2½ | 15 | 30 | 4 |
|---|---|---|---|
| days | contact hours | participants | expert lecturers |

DAY 1 · WEDNESDAY 18 NOVEMBER

# Foundations & wide-area assessment

## MORNING

| | | |
|---|---|---|
| 08:30 | Registration and tea | Badge collection, laptop check and Earth Engine login verification. |
| 09:00 | **Opening ceremony** | Welcome address, inauguration and course objectives. |
| 09:20 | **Module 1 · Foundations of remote sensing across scales** | Passive and active sensing; the resolution quartet; swath versus resolution; what “large-scale” changes; the observation continuum from satellites to ground surveys; Earth observation across the disaster management cycle. |
| 10:10 | **Module 2 · Data access and international emergency mapping** | Sentinel, Landsat, MODIS, VIIRS, GPM IMERG and global DEMs; Analysis-Ready Data; Copernicus Data Space and the Earth Engine catalogue; the International Charter, Copernicus EMS, Sentinel Asia and UN-SPIDER, and how a Pakistani agency requests activation. |
| 10:55 | Tea break | |
| 11:10 | **Module 3 · UAV systems, regulation and payloads** | UAV system composition; fixed-wing, multirotor and VTOL platforms; optical, multispectral, thermal and LiDAR payloads; regulations, authorities and operating procedures in Pakistan. |
| 12:00 | Lunch and prayers | |

DAY 1 · WEDNESDAY 18 NOVEMBER

# Mapping the flood & planning the flight

AFTERNOON

| | |
|---|---|
| 13:30 | **Lab 0 · Environment check**<br>Earth Engine Code Editor orientation, shared scripts, a first run and login troubleshooting. Offline data package for anyone without access. |
| 14:00 | **Lab 1A · Rapid flood mapping from Sentinel-1 SAR**<br>Pre- and post-event VV imagery, speckle filtering, change-detection ratio and thresholding, terrain masking with slope and HAND. Output: flood extent for the 2022 Sindh floods. |
| 15:30 | Tea break |
| 15:45 | **Lab 1B · Exposure and impact quantification**<br>Flood extent against WorldPop, ESA WorldCover and OpenStreetMap to build a district-wise exposure table; error sources and where they justify an aerial survey. |
| 16:20 | **Lab 2 · Flight planning and mission design**<br>Area, sensor, overlap, ground sample distance and safety; avoiding restricted airspace; altitude, speed and route generation. Each group plans the mission it flies on Day 2. |
| 17:00 | Close of Day 1<br>Field safety and pre-flight briefing issued. |

DAY 2 · THURSDAY 19 NOVEMBER

# Field operations & UAV data processing

| | | |
|---|---|---|
| 08:30 | **Transfer and pre-flight briefing** | To the NUST field site. Airspace, weather and safety limits, crew roles and go/no-go; aircraft checks and ground control layout. |
| 09:00 | **Field practical · UAV flight operations** | Supervised stations: manual handling, emergency take-off and landing, return-to-home; the planned mission; orthophoto, oblique and thermal acquisition; GCP survey. |
| 11:30 | **Data offload and quality check** | Coverage, overlap and image quality; flight log; transfer to workstations. |
| 12:00 | Lunch and prayers | Return to IGIS. |
| 13:30 | **Lab 3A · Photogrammetric processing** | POS data, pre-processing, alignment, dense reconstruction, DSM and orthomosaic, georeferencing and accuracy. |
| 15:15 | Tea break | |
| 15:30 | **Lab 3B · 3D modelling and damage quantification** | Oblique modelling; change detection between epochs; landslide displacement; building damage and road length, reconciled with the Lab 1 exposure table. |

*If weather or airspace prevents flying, a pre-flown reference dataset keeps the labs on schedule.*

DAY 3 · FRIDAY 20 NOVEMBER · HALF DAY

# Integration, case studies & rapid response

**09:00** **Module 4 · Integrating the observation chain**
Nested assessment from satellite screening to UAV tasking and ground validation; handover points, authority and latency; uncertainty and defensible figures.

**09:35** **Applied case studies**
Attabad Lake flood risk · Shisper Lake GLOF with HEC-RAS 2D · Jhang riverine flooding, ENSO and the monsoon.

10:15 Tea break
Scenario cards and data packages distributed.

**10:30** **Capstone · Rapid-response simulation**
Six groups, 45 minutes, a four-slide brief: event footprint, exposure table, UAV tasking plan and confidence statement. Scenarios include Dera Ghazi Khan floods, a Hunza GLOF, a Muzaffarabad earthquake, Tharparkar drought, Karachi urban flooding and Sindh reconstruction.

**11:45** **Communication, summary and closing**
Pak-NSDI and data sharing; self-study packages and online clinic; remarks by dignitaries, certificates and group photograph.

12:15 Close
Ahead of Friday congregational prayers.

LEARNING OUTCOMES

# What you'll be able to do.

01 Choose platform, sensor and resolution for a hazard and reporting scale.

02 Access free satellite archives and request international emergency mapping.

03 Map flood extent from Sentinel-1 SAR and quantify exposure.

04 Select a UAV platform and payload against mission requirements.

05 Follow Pakistan's UAV regulations, permissions and flight-safety procedures.

06 Plan and fly a survey mission for orthophoto, oblique and thermal data.

07 Process imagery to an orthomosaic, a DSM and an oblique 3D model.

08 Quantify damage and reconcile it with satellite exposure estimates.

09 Design a nested multi-scale workflow with stated uncertainty.

10 Deliver a rapid situation brief under time pressure.

EXPERT LECTURERS

# Your lecturers.

NY

## Dr. NIE Yong

Professor · Institute of Mountain Hazards and Environment (IMHE), CAS

Director, Mountain Science Data Center. Glacier and glacial lake hazards and remote sensing; 50+ papers including ESI highly cited work; risk assessment for the China–Pakistan Economic Corridor.

WW

## Engr. WANG Wen

Assistant Engineer · Mountain Science Data Center, IMHE, CAS

Remote sensing for mountain hazards and the cryosphere; extensive UAV fieldwork and processing, from flight planning to orthomosaics, DEMs and 3D models.



## Dr. Javed Iqbal

Professor · SCEE – IGIS, NUST

Three decades in GIS, remote sensing and hazard mapping of landslides, glacier avalanches and flash floods; 200+ scholarly contributions.

SK

## Dr. Shahid Nawaz Khan

Assistant Professor · SCEE – IGIS, NUST

Explainable machine learning for Earth observation, SAR–optical fusion, Earth Engine and UAV remote sensing; FAA Part 107 remote pilot.

BEFORE YOU ARRIVE

# Come ready to work.

The schedule is compressed, so there is no slack to recover from a failed login on the day.

- [ ] **Google Earth Engine access** registered and verified with a linked Cloud project. Allow 7–10 days.
- [ ] **Laptop with Google Chrome.**
- [ ] **QGIS 3.34 LTR** installed, with 5 GB free disk space.
- [ ] **UAV ground-station app** on a charged phone or tablet, with a cable.
- [ ] **Field clothing for Day 2:** closed-toe shoes, sun protection and a hat.
- [ ] **Pre-reading pack**, circulated two weeks ahead.
- [ ] **Signed field-safety acknowledgement**, handed in at registration.

VENUE & CONTACT

# Where to find us.

LECTURES & LABS · ALL THREE DAYS

## Institute of Geographical Information Systems

SCEE, NUST Sector H-12, Islamabad 44000

UAV FLIGHTS · DAY 2 MORNING

## NUST Field Site

Near the Saddle Club, NUST H-12 · transfer from IGIS at 08:30

**NUST_CPJRC_WRKSHP@igis.nust.edu.pk**
051 886 4479 (Dr. Shahid Nawaz Khan, NUST)

NOTES

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